

Village Campaign' are also promoting LED usage in street-lighting, informs Singh.

### Notable market trends

An interesting market trend which has been observed is that, the demand for India-made LED light bulbs is higher when compared to imported LED light bulbs. Raushan says, "Imported LED light bulbs do not have the electronics to withstand Indian electrical systems and, thus, do not work for very long. LED manufactured in India has good reliability and provides better after-sales services, which is not the case with imported LED light bulbs." He adds, "Although LED imported from China is cheaper, it is of bad quality and does not last long, and thus people prefer Indian LED light bulbs." On a similar note, Dhawan says, "Government should help the industry by restricting cheap, imported products that actually harm consumers as they are not in line with the quality and safety requirements."

### Challenges faced by the industry

- Lack of awareness regarding LEDs makes it difficult to scale-up business for the mass market
- Inefficient LEDs imported from China resulting in huge competition
- High cost of manufacturing; return on investment (ROI) made in tools in India is very low
- Lack of standards and LED testing facilities
- Heavy dependence on import of LED-related electronic components
- Lack of skilled manpower

Some other recent trends in the market include price reduction and shift toward manufacturing of low- and mid-power LEDs. LED lights are extensively used in automotive lighting, too. "The use of LED lamps in digital signage is another trend that has been observed," informs Lahiri.

As the LED street-lighting segment is becoming more mature, newer risk-sharing models are being explored between municipalities and lighting solution providers. Lalchandani says, "Some of these risk-sharing models comprise deferred payments, longer warranties, special-purpose investment entities for large investments

and annuity based payments." He adds, "Nowadays, many customers also seem to be looking up at lighting solutions companies as one-stop-shop providers, who can offer end-to-end solutions right from design, installation, testing, and commissioning and financing."

### LEDs are preferred over CFLs and tubelights for new installations

"We have observed that most people today, who are going for new installation in offices, homes, industries, factories and other commercial spaces, prefer to go for LEDs. It is slightly costly



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